Eigenvalue Distributions in Yang-Mills Integrals

(Summer Project Report)

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Abstract

In this project we have simulated zero-dimensional Yang-Mills matrix models with gauge group $SU(N)$ using Monte Carlo method. We computed the eigenvalue distributions of the model and fitted them with Wigner semicircle. Our simulation results are in good agreement with the analytical results.

Acknowledgements

I owe my sincere gratitude to Dr. Anosh Joseph, Project Supervisor at IISER Mohali for providing me an opportunity to do the project work and giving me all support and guidance, which made me to complete the project duly. My special thanks to my Mr. Navdeep Singh Dhindsa who helped me in every difficulties related to C++ programming I faced.

I would like to thank the institute, IISER Mohali, for giving this opportunity. I would like to thank everyone involved in organizing this Summer Program at IISER Mohali.

Last but not the least, I would like to expand my thanks to everyone who directly or indirectly helped me in completing this project.

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1 Introduction

In the recent times there has been renewed interest in dimensional reductions of $SU(N)$ Yang-Mills theories. We can use these "toy models" to recover the full unreduced theories in the large N limit. The supersymmetric reductions are relevant to D-brane physics and have also been used in various attempts to define nonperturbative formulations of quantum gravity, supermembranes and superstrings. For the complete reduction, all spacetime dependence is eliminated from the gauge "fields," and the continuum Yang-Mills path integral becomes an ordinary multi-matrix integral [1].

$$\begin{array}{rcl} \text{matrix field theory} & \simeq & \text{suitable zero dimensional matrix model} \\ (N=\infty) & \simeq & (N=\infty). \end{array}$$

That is, it appears that every matrix field theory can be replaced by a suitable matrix integral such that at least some of the physical observable on both sides become identical at $N = \infty$.

In [2, 3] and [4], it is uncovered that in many cases of interest these matrix integrals converge and thus do not need to be regulated. Mathematically rigorous proofs for $N > 2$ are still missing.

2 Yang-Mills integrals

The integrals in question are

$$\begin{aligned} \mathcal{Z}^{\mathcal{N}}_{D,N} &= \int \prod_{A=1}^{N^2-1} \left(\prod_{\mu=1}^{D} \frac{dX^A_\mu}{\sqrt{2\pi}} \right) \left(\prod_{\alpha=1}^{\mathcal{N}} d\Psi^A_\alpha \right) \\ & \quad \exp\left[\frac{1}{2} Tr\left[X_\mu, X_\nu\right]\left[X_\mu, X_\nu\right] + Tr\Psi_\alpha \left[\Gamma^\mu_{\alpha\beta} X_\mu, \Psi_\beta\right] \right]. \end{aligned} \tag{1}$$

Above equation correspond to fully reduced D-dimensional Euclidean $SU(N)$ Yang-Mills theory. Here $\mathcal{N}$ is the number of real supersymmetries.

1. For $\mathcal{N} > 0$ the only possible dimensions are $D = 3, 4, 6, 10$ corresponding to $\mathcal{N} = 2, 4, 8, 16$, respectively.

2. For $\mathcal{N} = 0$ (no supersymmetry), a priori all $D \geq 2$ are possible

Here we omit the terms with Grassmann fields both from the measure and action of Eq.(1).

The necessary and sufficient conditions of existence for the integrals Eq. (1) are:

$$D = 4, 6, 10 \quad \text{and} \quad N \geq 2 \quad \text{for} \quad \mathcal{N} > 0 \tag{2}$$

For $\mathcal{N} = 0$

$$\begin{aligned} D = 3 &\quad \text{and} \quad N \geq 4 \\ D = 4 &\quad \text{and} \quad N \geq 3 \\ D \geq 5 &\quad \text{and} \quad N \geq 2 \end{aligned} \tag{3}$$

The distribution for the eigenvalues λ_i of just one matrix, say X_1, in the background of the other matrices $X_2, ..., X_D$

$$\rho\,(\lambda) = \frac{1}{N}\left\langle \sum_{i=1}^{N} \delta(\lambda - \lambda_i) \right\rangle. \tag{4}$$

Here, the average $\langle \cdots \rangle$ is with respect to Eq. (1).

In an ordinary Hermitian matrix model, the density Eq. (4) falls off at infinity as $\rho(\lambda) \sim \exp(-\lambda^2)$.

The most divergent contribution is not obtained by simply counting the overall dimensionality of the integral. In fact, a more dangerous infrared configuration stems from only a single (say, the i-th) eigenvalue x^i_μ becoming large in the D-dimensional space.

Numerical estimate of the eigenvalue spectrum of X_1 by generating ensembles of $X = (X_1, X_2, \cdots, X_D)$ with the statistical weight

$$Z^{\mathcal{N}}_{D,N}\,(X) = \exp\left[\frac{1}{2}\mathrm{Tr}\,[X_\mu, X_\nu]\,[X_\mu, X_\nu]\right]\mathcal{P}_{D,N}\,(X) \tag{5}$$

where $\mathcal{P}_{D,N}\,(X)$ is the Pfaffian polynomial coming from the integration over the fermionic degrees of freedom. (It is simply dropped for the bosonic integrals.)

In the Yang-Mills case the asymptotic behavior of the one-matrix eigenvalue distribution Eq. (4) decays algebraically

$$\rho\,(\lambda) \sim |\lambda|^{-\alpha} \qquad \text{for} \qquad \lambda \to \pm\infty, \tag{6}$$

where

$$\alpha = \begin{cases} 2D - 5 & \text{for} \quad \mathcal{N} > 0 \\ 2N(D-2) - 3D + 5 & \text{for} \quad \mathcal{N} = 0 \end{cases} \tag{7}$$

The bosonic case is much more conventional in that the density becomes concentrated in a finite interval at large N.

2.1 Bosonic integrals

In this project, we are interested in stimulating Yang-Mills integrals with no supersymmetry (bosonic case).

$$\mathcal{Z}_{D,N} = \int \prod_{A=1}^{N^2-1} \left(\prod_{\mu=1}^{D} \frac{dX_\mu^A}{\sqrt{2\pi}} \right) \exp\left[\frac{1}{2} \mathrm{Tr}\left[X_\mu, X_\nu\right]\left[X_\mu, X_\nu\right] \right]. \tag{8}$$

Note that gauge fixing is no longer required here, since the over counting of gauge-equivalent configurations involves merely a factor of the compact, finite volume of the gauge group: spacetime has become a point (or more precisely, an infinitesimal torus, since the "point" still keeps a sense of the D directions).

The Euclidean action is

$$S\left(X\right) = -\frac{1}{2}\ \mathrm{Tr} \sum_{\mu=1,\nu=1}^{D} \left[X_\mu, X_\nu\right]\left[X_\mu, X_\nu\right]. \tag{9}$$

3 Monte Carlo method

Let us consider an integral in one dimension, of a well-defined function $f(x)$ in the interval x_i and x_f

$$I = \int_{x_i}^{x_f} f(x)dx. \tag{10}$$

If n random values are generated between lower limit x_i and upper limit x_f, then average value of f over this interval can be given by

$$\bar{f}_n = \frac{1}{n}\sum_{r=1}^{n} f(x_r). \tag{11}$$

Thus above integral is given by

$$I = \int_{x_i}^{x_f} f(x)dx \simeq (x_f - x_i)\bar{f}_n = (x_f - x_i)\frac{1}{n}\sum_{r=1}^{n} f(x_r). \tag{12}$$

This method is called the Monte Carlo (MC) method.

3.1 Monte Carlo with importance sampling

The method of importance sampling tries to increase the efficiency of the Monte Carlo method by choosing a function that is more flat. The random numbers, x_r, are selected according to a probability density (or weight) function, $w(x)$, which is normalized to unity

$$\int_{x_i}^{x_f} w(x)dx = 1. \tag{13}$$

We have the integral

$$I = \int_{x_i}^{x_f} w(x)f(x)dx \simeq \frac{1}{n}\sum_{r=1}^{n} f(x_r). \tag{14}$$

In order to reduce the Monte Carlo error we can either increase the sample size N or decrease the variance σ. We can reduce the variance by writing the above integral as

$$I = \int_{x_i}^{x_f} w(x)\frac{f(x)}{w(x)}dx. \tag{15}$$

If we have set of N random numbers x_r, which are distributed according to the $w(x)$, then the above integral can approximately be written as

$$I_N = \frac{1}{N}\sum_{r=1}^{N}\frac{f(x_r)}{w(x_r)}. \tag{16}$$

Note: $w(x)$ is chosen such that $\frac{f(x_r)}{w(x_r)}$ slowly varying, since this will reduce the standard deviation.

3.2 Markov chains

Markov chain Monte Carlo (MCMC) is also a random sampling method. It is used to visit a point x with a probability proportional to some given distribution function say, $\pi(x)$. MCMC "automatically" puts its sample points preferentially where $\pi(x)$ is large, in direct proportion. In MCMC, we sample distribution $\pi(x)$ by a Markov chain. A Markov chain is a sequence of points, $x_0, x_1, x_2, \cdots$, that are locally correlated, with the ergodic property. The sequence of points will eventually visit every point x, in proportion to $\pi(x)$. In a Markov chain, each x_i chosen from a distribution depends only on the value of the immediately preceding point x_{i-1}. The chain has memory extending only to one previous point and it is completely defined by a transition probability function of two variables, $p\left(x_i|x_{i-1}\right)$.

If $p\left(x_i|x_{i-1}\right)$ is chosen to satisfy the detailed balance equation

$$\pi\left(x_1\right) p\left(x_2|x_1\right) = \pi\left(x_2\right) p\left(x_1|x_2\right), \tag{17}$$

then the Markov chain will in fact sample $\pi(x)$ ergodically.

If $\{X_n\}_{n\geq 0}$ is a Markov Chain with the initial distribution $\pi(x)$ and transition matrix T if for all $n \geq 0$

1. $P(X_0 = a) = \pi(a)$,
2. $P\left(X_{n+1} = a|X_0 = b_1, \cdots, X_n = b_n\right) = P\left(X_{n+1} = a|X_n = b_n\right) = T\left(a, b_n\right)$.

3.3 Metropolis algorithm

It is used to obtain a chain of random numbers from a probability distribution. It can get random samples from any probability distribution $\Pi(x)$, given that (i.) we have a function $\pi(x)$ in hand, which is proportional to the density of Π, i.e., $\pi(x) = \frac{1}{Z}\Pi(x)$, and (ii.) the values of $\pi(x)$ can be calculated.

Note: Computing the required normalization factor Z is often extremely difficult.

The Metropolis algorithm for our system is as follows

1. Choose the initial field configuration say $\{X_n\}_{n\geq 0}$. Random field configuration is known as hot start and if all the field configuration is set to 0, then it is known as cold start.
2. Then choose any random direction in field configuration space and make a jump in field configuration and lets call this x_{new}.
3. Then compute $\Delta S = S\left[x_{new}\right] - S\left[x_{old}\right]$.
4. If $\Delta S \leq 0$, then accept this field configuration.
5. If $\Delta S > 0$, then choose a random number r in the interval $[0, 1]$ and if $r < e^{-\Delta S}$ then accept the field configuration, otherwise reject it.
6. Repeat step 2) to 5) for many times. This repetition of the steps depends on the how accurately we want our results to be. The thermalization should lie in the beginning part of these steps.
7. Calculate the average of the observable from the field configuration after the thermalization process.

4 Analytical results

When we consider the pure Yang-Mills term only as in Eq.(9) the average value of the action is given by

$$\frac{\langle S \rangle}{N^2 - 1} = \frac{d}{4}, \tag{18}$$

where N is the order of traceless Hermition matrices and d is the dimension of the spacetime before compactification.

5 Simulation results

The form of action for bosonic Yang-Mills considered in this stimulation is

$$S_{YM}\left[X\right] = -N \sum_{\mu=1}^{d} \sum_{\nu=\mu+1}^{d} \left(X_\mu X_\nu X_\mu X_\nu - X_\mu^2 X_\nu^2\right), \tag{19}$$

where X_μ's are $N \times N$ Hermition traceless matrices.

Wigner semicircle distribution

$$f(x) = \frac{2}{\pi R^2} \sqrt{R^2 - x^2} \tag{20}$$

for $-R \leq x \leq R$, and $f(x) = 0 \quad if \quad |x| > R$.

For the simulation to be efficient, and for the result to be considerably accurate, the jump parameter is set such that acceptance rate is between 60% *to* 80%.

We have generated 5 plots for each ($d = 24, N = 2$), ($d = 32, N = 3$), ($d = 45, N = 4$), ($d = 16, N = 5$), ($d = 28, N = 6$), and ($d = 70, N = 8$).

1. In Figs. (1), (6), (11), (16), (21), and (26) we have shown the plots of acceptance rate against Monte Carlo time. To calculate the acceptance rate we have divided the total sweeps in gaps of either 500 (for $d = 24, N = 2$) or 1000 (for the rest) and calculated the acceptance rate by formulae (sweeps $-$ reject) $\times$ 100.0/sweeps, where sweeps is size of gap.

2. In Fig. (2), (7), (12), (17), (22), and (27) we have shown the plots of the thermalization of the action for different d and N values against Monte Carlo time. In these plots we have shown the value of action at gap of 500 (for $d = 24, N = 2$) or 1000 (for the rest). In these we have also zoomed in the figure, and find that average value of action is in excellent agreement with theoretical value of action given in Eq. (18).

3. In Figs. (3), (8), (13), (18), (23), and (28) we have shown the plots of fractions of eigenvalues in different interval of the eigenvalues. To plot these graphs we first searched for the maximum and the minimum eigenvalues, then then divided the interval between the maximum and the minimum eigenvalues into smaller bins. After this, we counted the number of eigenvalues that fell in each bin interval and then divided the number of eigenvalues in each bin interval with the total number of eigenvalues.

4. In Figs. (4), (9), (14), (19), (24), and (29) we have shown the plots of the eigenvalue distributions. In these plots on the y-axis we have plotted the probability density and on the x-axis we have plotted the eigenvalues.

5. In Figs. (5), (10), (15), (20), (25), and (30) we have fitted the eigenvalue distribution plots with the Wigner semicircle.

Note: For Figs. of points (3), (4), and (5), we have taken X_μ after thermalization to calculate the eigenvalues.

5.1 Plots for $N = 2, d = 24$

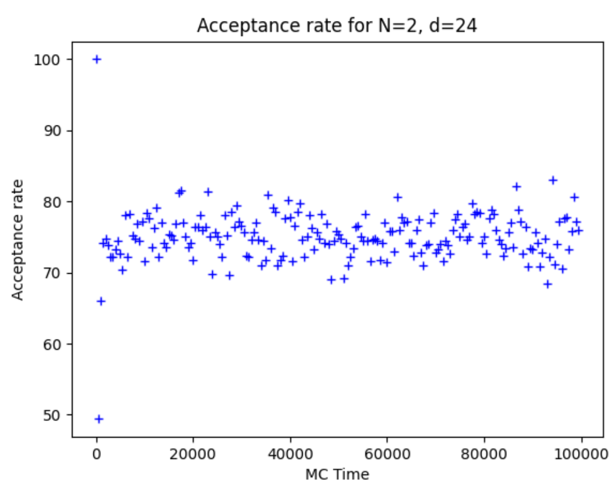
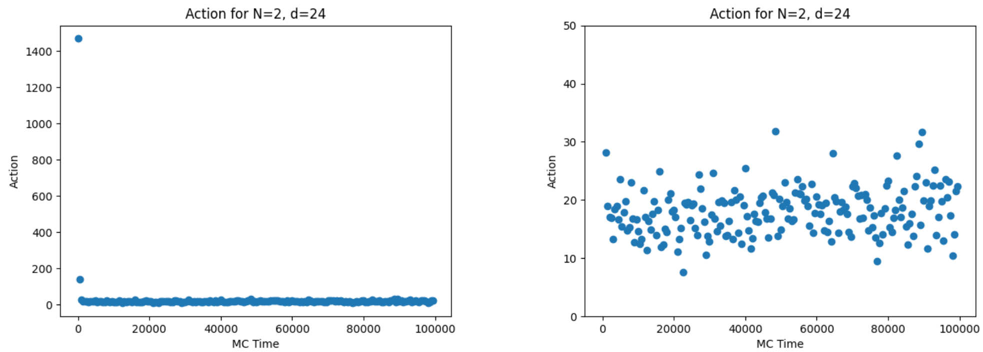


Figure 1: Acceptance rate against Monte Carlo time.



(a) (b)

Figure 2: (a): Plots showing the thermalization of S with Monte Carlo time (b): Zoomed version of (a).

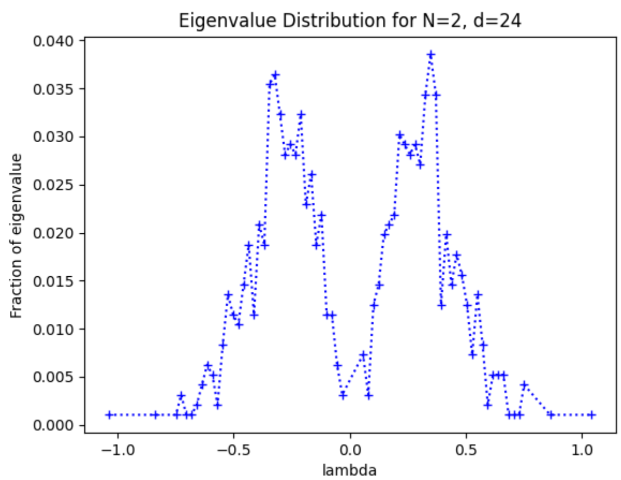


Figure 3: Fraction of the eigenvalues against the eigenvalues.

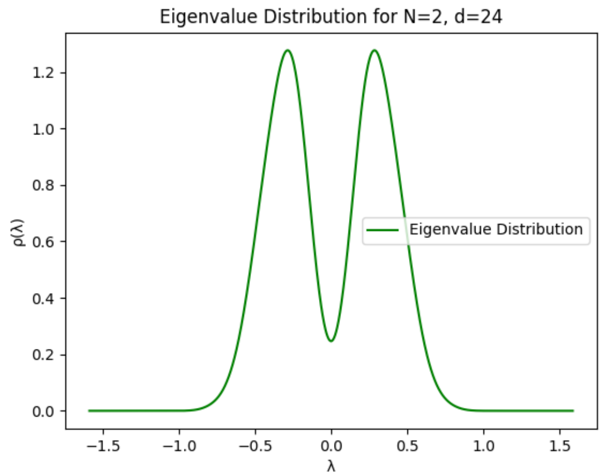


Figure 4: Eigenvalue distributions.

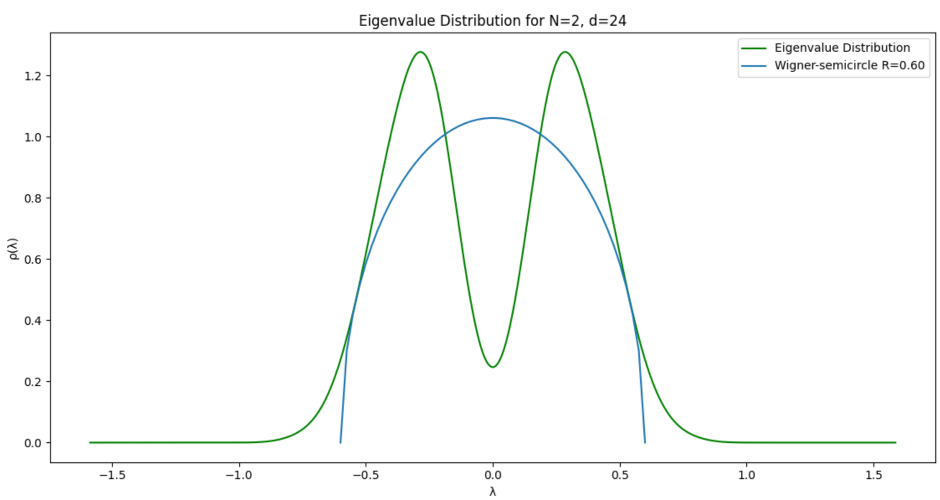


Figure 5: Eigenvalue distribution fitted with Wigner semicircle.

5.2 Plots for $N = 3, d = 32$

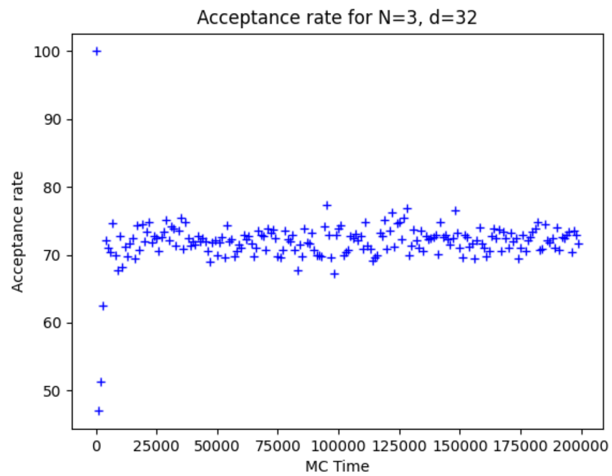
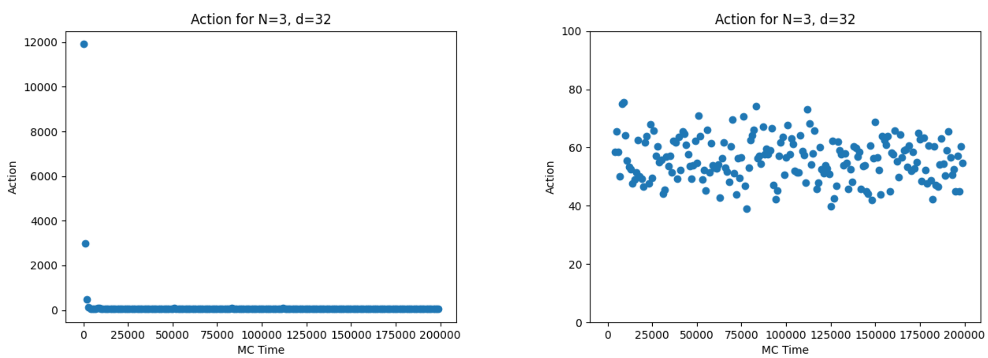


Figure 6: Acceptance rate against Monte Carlo time.



(a)

(b)

Figure 7: (a): Plots showing the thermalization of S with Monte Carlo time (b): Zoomed version of (a).

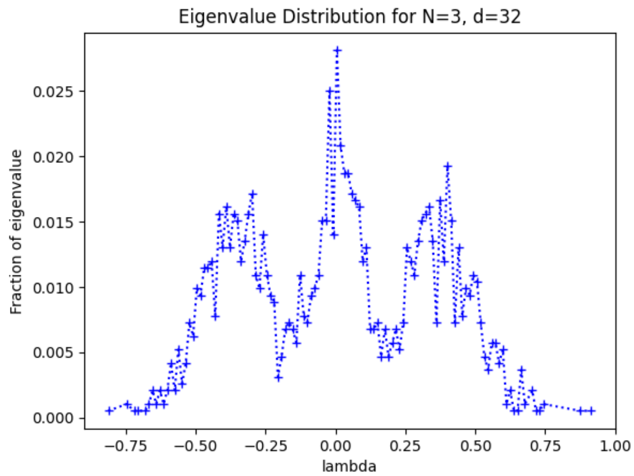


Figure 8: Fraction of the eigenvalues against the eigenvalue.

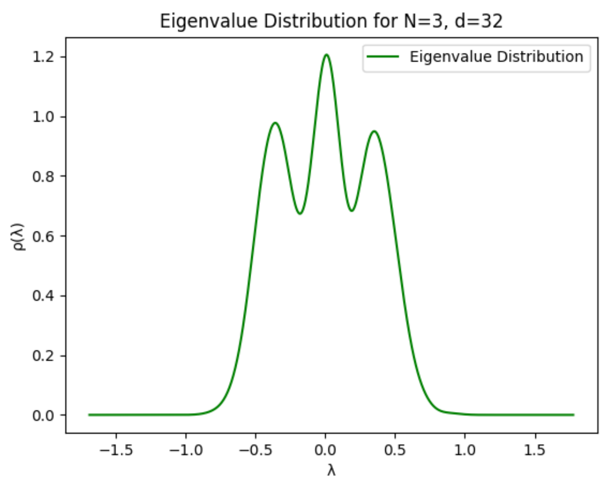


Figure 9: Plots of the eigenvalue distributions.

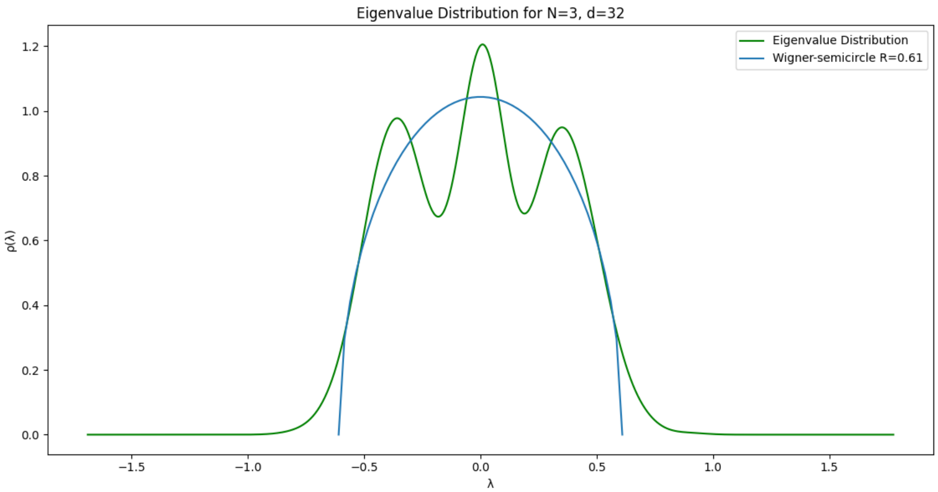


Figure 10: The eigenvalue distribution plots fitted with the Wigner semicircle.

5.3 Plots for $N = 4, d = 45$

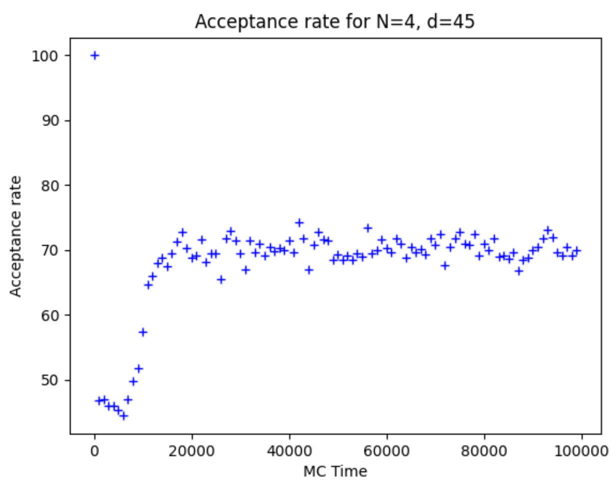
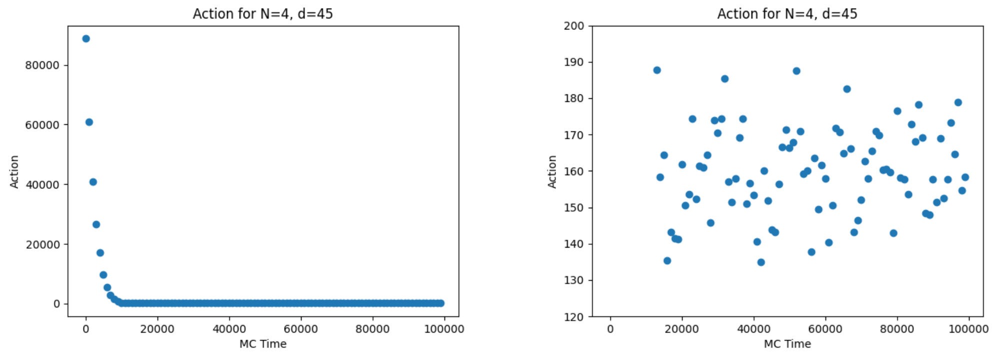


Figure 11: Acceptance rate against Monte Carlo time.



(a) (b)

Figure 12: (a): Plots showing the thermalization of S with Monte Carlo time (b): Zoomed version of (a).

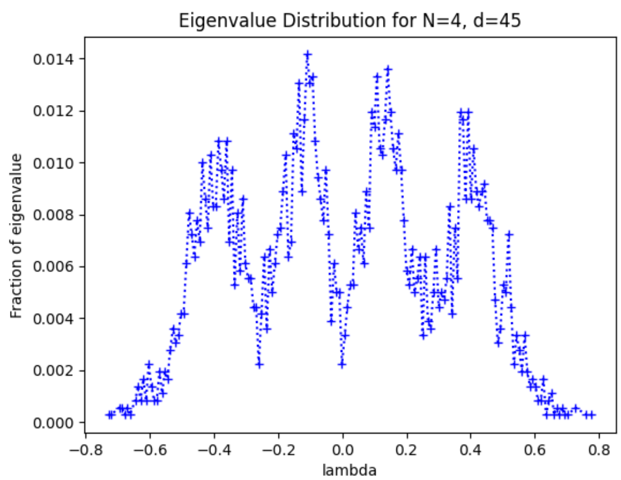


Figure 13: Fraction of the eigenvalues against the eigenvalues.

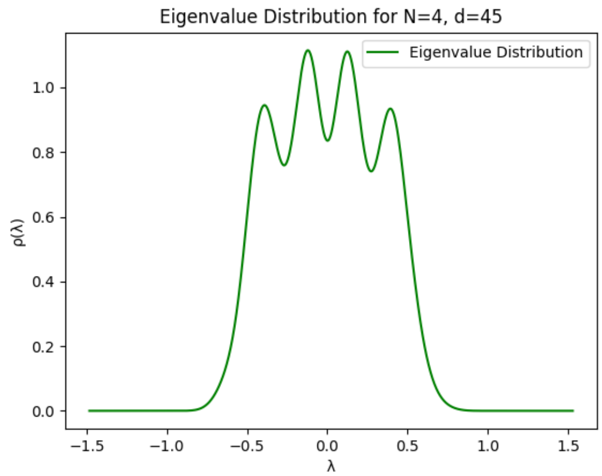


Figure 14: Plots of the eigenvalue distributions.

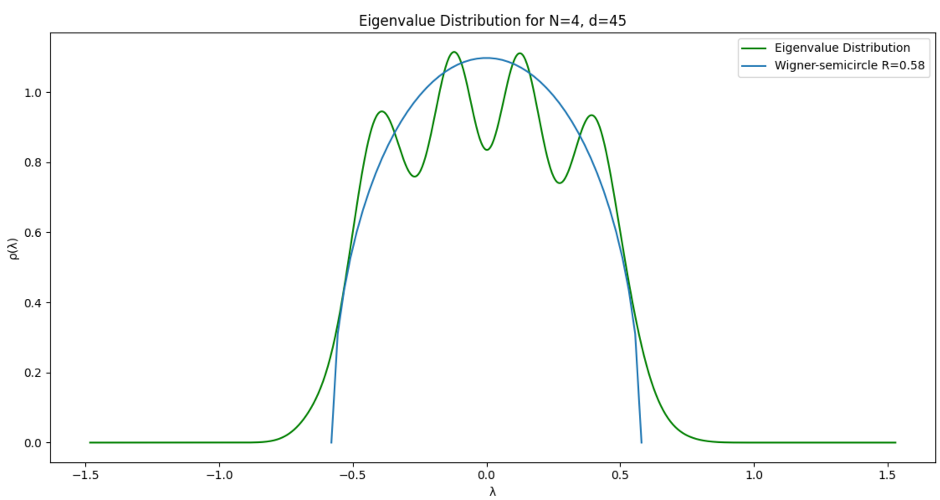


Figure 15: Eigenvalue distribution plots fitted with the Wigner semicircle.

5.4 Plots for $N = 5, d = 16$

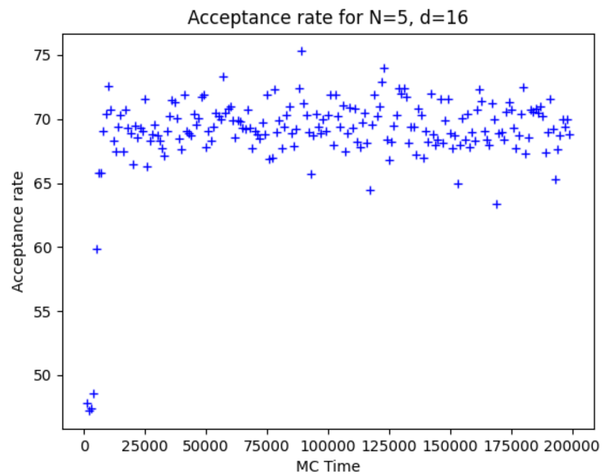
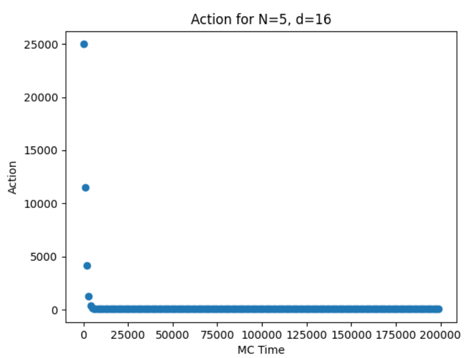
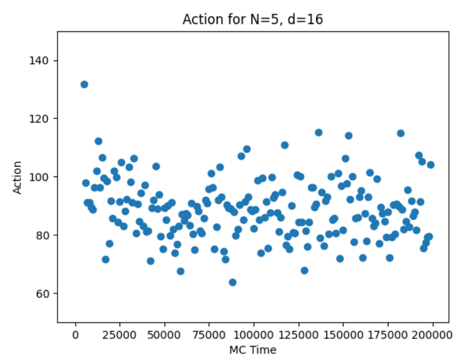


Figure 16: Acceptance rate against Monte Carlo time.



(a)



(b)

Figure 17: (a): Plots showing the thermalization of S with Monte Carlo time (b): Zoomed version of (a).

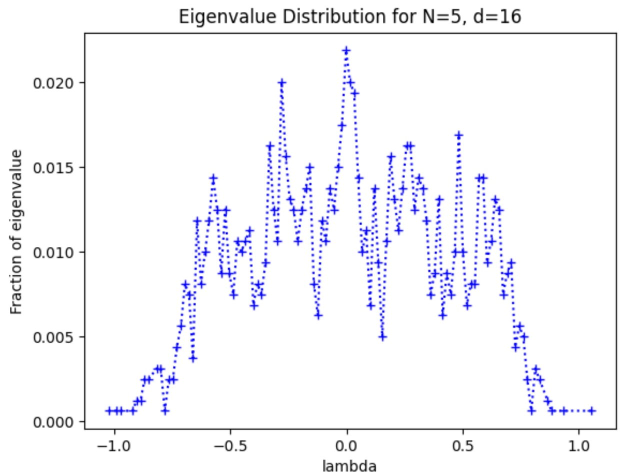


Figure 18: Fraction of the eigenvalues against the eigenvalues.

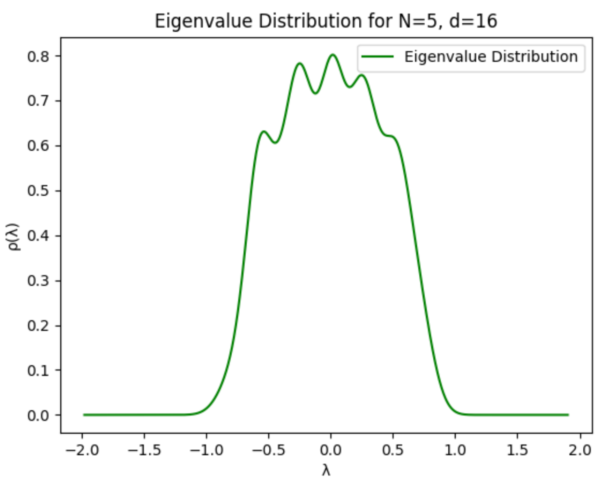


Figure 19: Plots of the eigenvalue distributions.

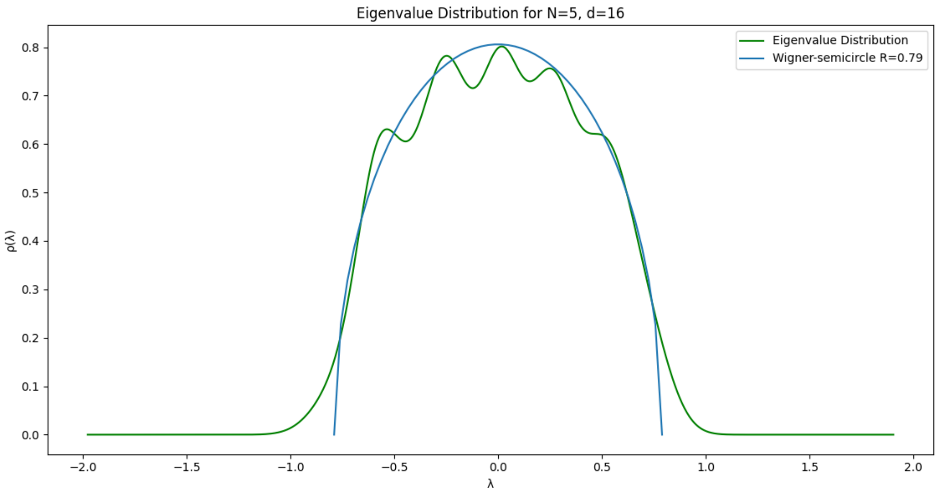


Figure 20: Eigenvalue distribution plots fitted with the Wigner semicircle.

5.5 Plots for $N = 6, d = 28$

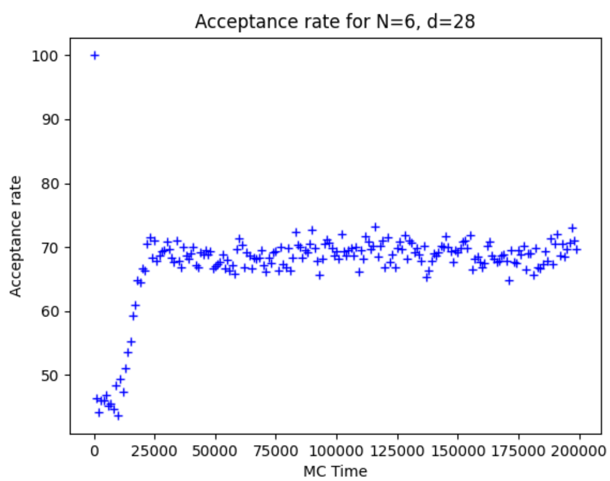
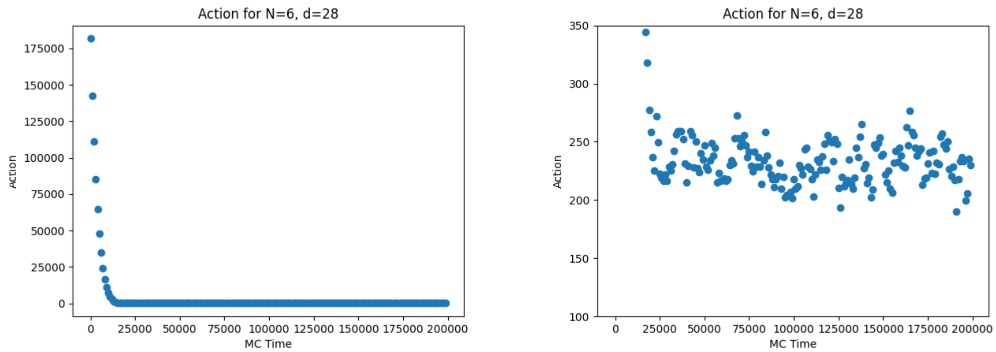


Figure 21: Acceptance rate against Monte Carlo time.



(a) (b)

Figure 22: (a): Plots showing the thermalization of S with Monte Carlo time (b): Zoomed version of (a).

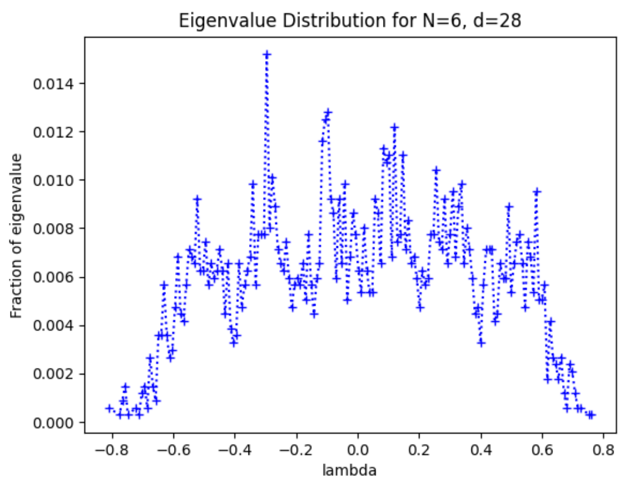


Figure 23: Fraction of the eigenvalues against the eigenvalues.

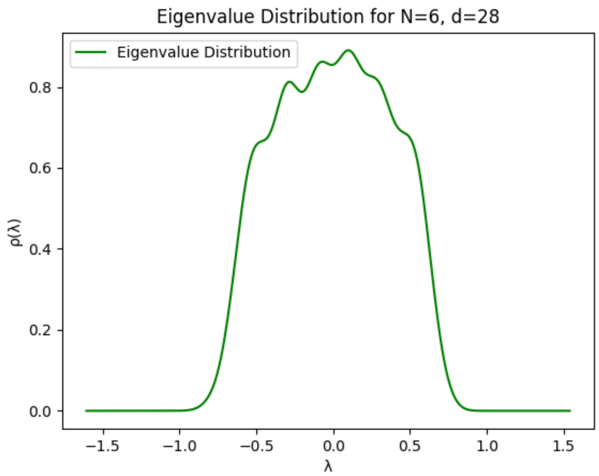


Figure 24: Plots of the eigenvalue distributions.

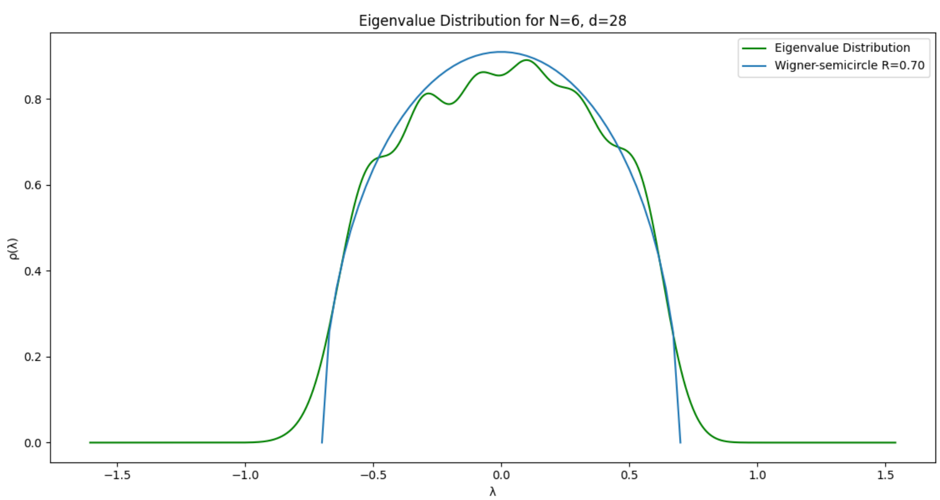


Figure 25: Eigenvalue distribution plots fitted with the Wigner semicircle.

5.6 Plots for $N = 8, d = 70$

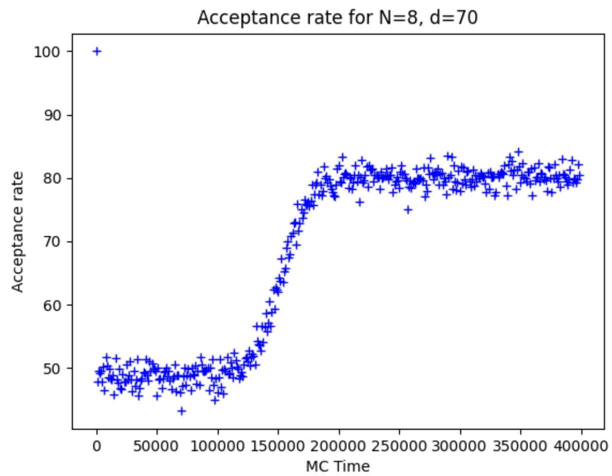
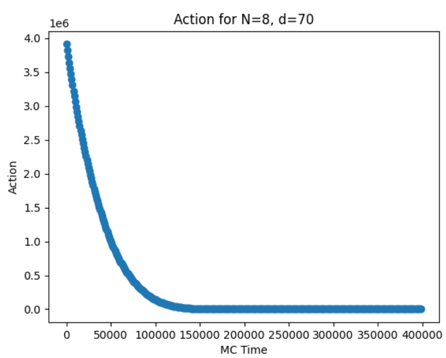
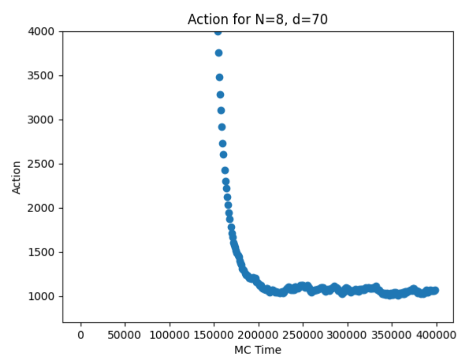


Figure 26: Acceptance rate against Monte Carlo time.



(a)



(b)

Figure 27: (a): Plots showing the thermalization of S with Monte Carlo time (b): Zoomed version of (a).

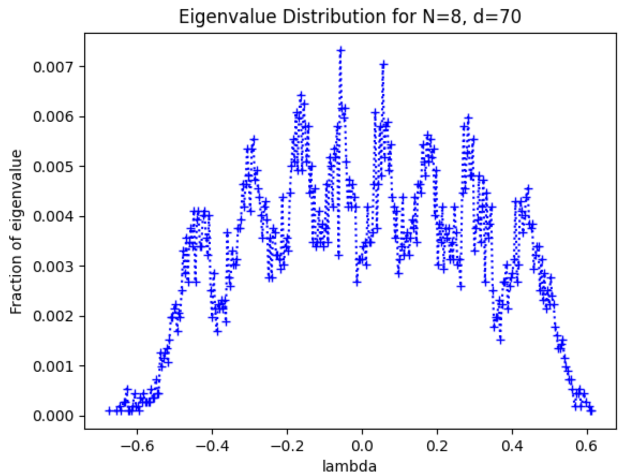


Figure 28: Fraction of the eigenvalues against the eigenvalues.

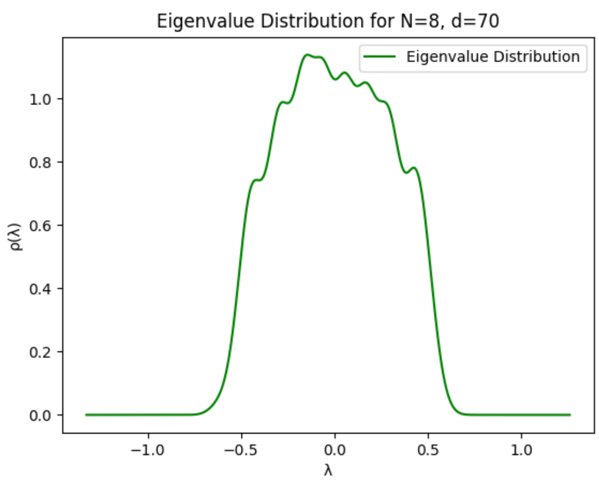


Figure 29: Plots of the eigenvalue distributions.

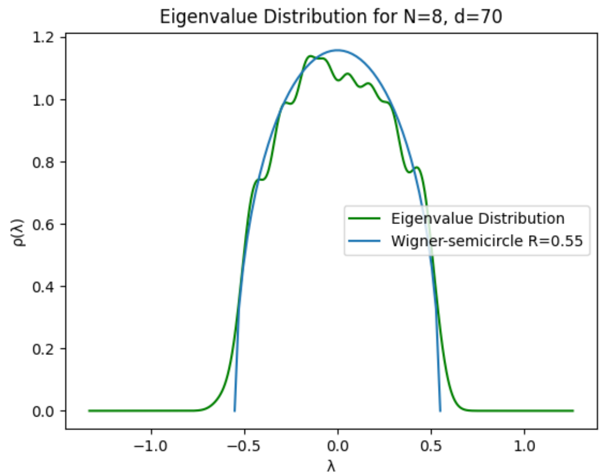


Figure 30: Eigenvalue distribution plots fitted with the Wigner semicircle.

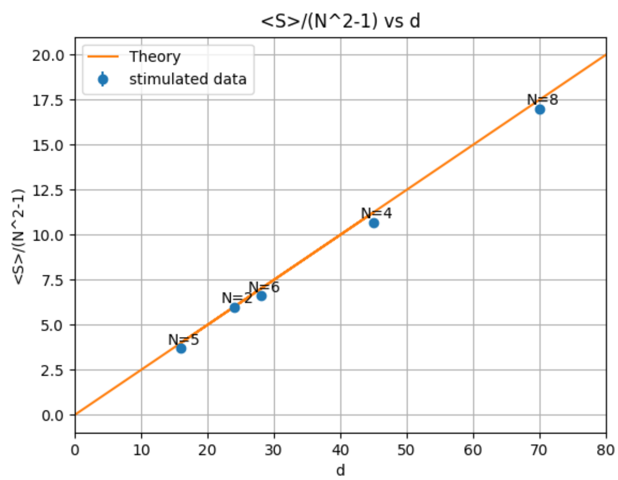


Figure 31: Combined plot of $\frac{\langle S\rangle}{N^2-1}$ against d for different values of N.

From Fig. (31) we can see that action is in excellent agreement with analytical value given in Eq (18).

From Figs. (4), (9), (14), (19), (24), and (29), we can see that eigenvalues of $X_\mu' s$ for any N and d are within the range of $(-\pi, \pi)$.

6 Discussions and conclusions

In this project we have simulated the zero-dimensional Yang-Mills matrix model using the method of Markov Chain Monte Carlo. We have plotted action value against MC time. From Fig. (31) we find that action is in excellent agreement with analytical value given in Eq.(18). From the eigenvalue distribution plots we find that the eigenvalues of $X_\mu' s$ for any N and d are within range of $(-\pi, \pi)$ and the number of peaks is equal to the order of matrices, i.e., N. From these plots we can also see that the width of the eigenvalue distributions becomes narrower as d increases.

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